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A Multimodal Approach with Explainable AI for Parkinson's Disease Stage Classification

Submitted by

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1 Introduction

Parkinson's disease (PD) is a worldwide prevalent progressive neurodegenerative disorder, resulting from the loss of dopaminergic neurons in the brain. The disease in a majority of cases occurs among old people, but people of younger age can also get infected. More men than women are infected. The cause of PD remains unidentified, but individuals with a familial history of the disease are at increased risk. In 2019, world estimates were over 8.5 million people with PD. Current estimates suggest that, in 2019, PD contributed to 5.8 million disability adjusted life years (DALYs), an increase of 81% since 2000, and resulted in 329 000 deaths, an increase of over 100% since 2000 [9]. According to recent data research conducted by the Statista Research Department, there are over 830,000 people with PD in western Europe, including 120,000 in France [6]. However, diagnosing PD remains challenging, as the disease shares symptoms with other neurodegenerative conditions, and current methods often rely heavily on clinical evaluation and the patient's response to treatment, making early diagnosis difficult. Early detection of Parkinson's disease remains challenging, and leveraging diverse datasets with advanced machine learning techniques can significantly improve classification accuracy [2]. SPECT imaging shows promise for early Parkinson's disease detection, but interpretation is complex and prone to human error. Deep learning models can automate PD detection with high accuracy, yet their "black box" nature limits clinical adoption. To overcome this, explainable AI approaches integrate CNN classifiers with interpreters, highlighting brain regions influencing predictions and providing transparent support for clinicians.

1.1 PD stage:

Clinicians utilize clinical rating scales to elucidate the PD movement and non-movement symptoms, their severity, and their effects on an individual's everyday activities. Clinical scales enable physicians to monitor the advancement of PD and are utilized in clinical trials. Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) An updated and enlarged version of the UPDRS, released in 2008. It is a more thorough scale created to assess the different aspects of PD. The four components included in this scale are [5]:

1. Non-movement aspects of experiences of daily living
2. Movement aspects of experiences of daily living
3. Movement examination
4. Movement complications

Table 1: Five Stages of PD Progression

Stage	**Description**
Stage 1	Mild symptoms that have no impact on daily activities. Tremor and movement symptoms on one side of the body. Postural, gait, and facial changes are possible.
Stage 2	Symptoms worsen with tremor, stiffness, and movement symptoms in both sides of the body. Activities of daily living become harder to do and take longer to complete.
Stage 3	Mid-stage with loss of balance and unsteadiness. There are increased falls. The motor symptoms worsen but the person can still sustain an independent existence with some constraints of physical ability.
Stage 4	Symptoms are maximally expressed and highly disabling. The patient cannot walk and stand independently but can continue living at home with caregiver help for activities of daily living.
Stage 5	Most critical and disabling stage. Rigidity hinders walking or standing. Person is bedridden or confined to a wheelchair and requires total care.

2 Motivation

Parkinson's disease (PD), which affects over 10 million individuals globally, is the second most prevalent neurodegenerative disorder following Alzheimer's disease. Each year, approximately 60,000 people are diagnosed for the first time in the United States alone, at an average age of starting at around 60 years. However, nearly 4% of cases occur before the age of 50, indicating that younger populations are also at risk. Traditional diagnostic approaches are based mainly on subjective clinical observations and patient questionnaires. Although useful, these methods may not detect early stage PD, missing nearly 25–30% of cases. Such diagnostic delays significantly hinder timely medical intervention and increase the likelihood of complications, such as Parkinson's dementia, which impacts about 30% of PD patients. Given these challenges, there is a critical need for improved diagnostic techniques that can enable earlier and more accurate identification of PD. Leveraging data-driven methods and advanced computational techniques offers a promising pathway to overcome the limitations of traditional approaches, ultimately contributing to better disease management and improved patient quality of life.

3 Specific Objectives and Possible Outcomes

3.1 Objectives

- To build a multimodal Parkinson's detection framework using MRI images and clinical data for more robust predictions.
- To improve generalization and interpretability via transfer learning (NTUA $\rightarrow$ PPMI) and XAI for clinically meaningful insights.

3.2 Possible Outcomes

1. Timely and precise diagnosis of Parkinson's disease.
2. Clinically interpretable predictions supporting decision-making.

4 Background and Present State

Various machine learning and deep learning methods have been extensively investigated for the detection of Parkinson's disease (PD). Researchers have explored both imaging-based methods, such as MRI and SPECT, and clinical or behavioral data to improve prediction accuracy. Recent trends focus on multimodal frameworks and explainable AI to enhance interpretability and clinical adoption.

To address feature interpretability, Abbas et al. [1] proposed C3BAM-XAI, a CNN enhanced with Convolutional Block Attention Module (CBAM) which integrates channel and spatial attention. That method highlighted critical MRI features relevant to PD detection, improving transparency of the classification process. Still, the work only utilized MRI data and did not integrate complementary clinical or behavioral information that could strengthen predictions.

Moving toward multimodal frameworks, Dentamaro et al. [3] introduced a deep learning model that fused MRI scans with clinical data using a novel Excitation Network (EN). Their results demonstrated improved performance compared to unimodal approaches, showing the promise of combining heterogeneous data sources. Nevertheless, the dataset used was small (90 patients from PPMI), which raises concerns about overfitting and limits the generalizability of the findings.

Esan et al. [4] used a Kaggle data set (n = 2105) containing demographics, medical history, lifestyle, cognitive assessments (UPDRS, MoCA), and symptoms reported by the clinician and the patient. The data were pre-processed with MinMax scaling, SMOTE for class balancing, and Sequential Backward Elimination for feature selection. Five ML models (SVM, KNN, LR, RF, XGBoost) and a stacked ensemble were evaluated, with Random Forest performing best. SHAP and LIME provided global and local interpretability.

Limitations included reliance on synthetic data, heterogeneous sources, lack of external validation, and lack of imaging data, restricting multimodal analysis.

Li et al. [7] proposed PIDGN, an explainable multimodal deep learning framework that fuses genetic (SNP) and MRI data for early Parkinson's disease prediction. The model uses EnsembleTree, 3D ResNet, and gated attention fusion, with SHAP and Grad-CAM providing interpretability. It achieved high accuracy but is computationally intensive, relies on a specific dataset, and does not include other modalities such as clinical or behavioral data.

Noun et al. [8], the authors designed a 3D Convolutional Neural Network (CNN) that captures spatial context from DaTSCAN SPECT images, achieving high accuracy in distinguishing healthy controls from PD patients at stages 1–3. However, this study relied solely on image data, lacked multimodal comparison with clinical information, and its black-box nature limited clinical interpretability.

In our study, we will develop a multimodal approach for Parkinson's disease stage classification by integrating multiple datasets and employing advanced algorithms. Our method will emphasize explainable AI to provide interpretable predictions, with the aim of improving early detection and supporting clinical decision-making.

5 Outline of Methodology

In this research, we propose a multimodal deep learning framework that integrates volumetric MRI data with clinical information for Parkinson's disease (PD) stage classification. The overall workflow is illustrated in Figure 1.

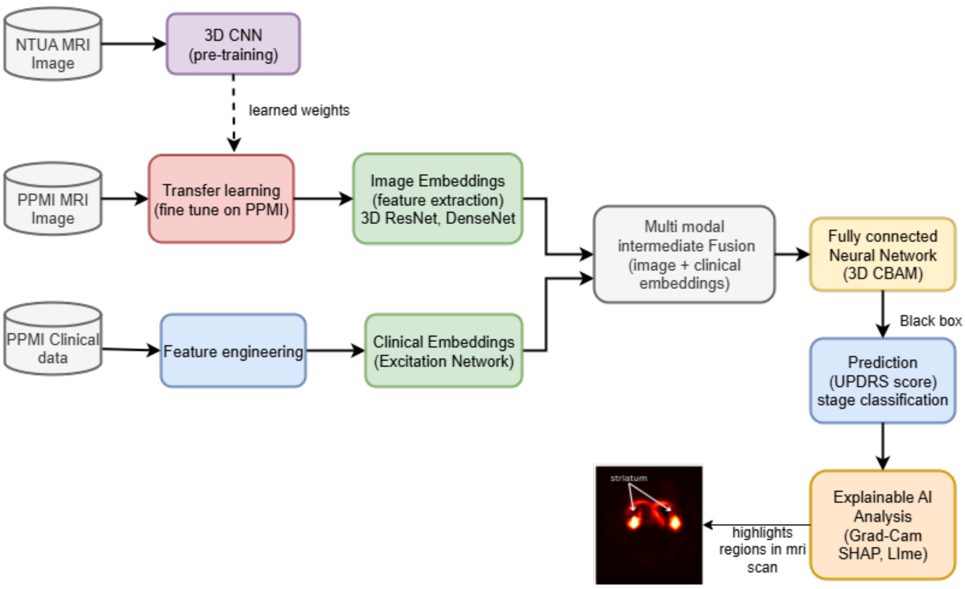

Figure 1: Work-flow diagram of PD stage classification

5.1 Dataset Acquisition

The MRI datasets used in this study were obtained from two sources:
NTUA MRI Images: The dataset was sourced from the National Technical University of Athens (NTUA) and consists of two classes: PD (55 patients) and Non-PD (23 patients). The total number of MRI images in this data set is 43,087, including 32,706 images of patients with PD and 10,381 images of patients without PD. An NTUA MRI image is shown in Fig. 2
PPMI MRI Images: The dataset was obtained from the Parkinson's Progression Markers Initiative (PPMI). It also includes two classes: PD (1086 patients) and Non-PD (270 patients), totaling 1,356 MRI images. A PPMI MRI image is shown in fig 3

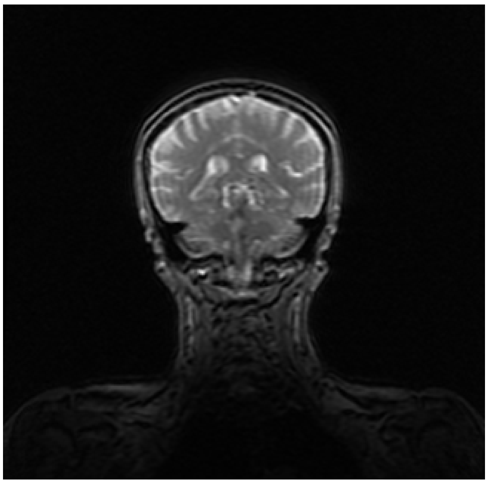

Figure 2: A MRI image from NTUA datasets

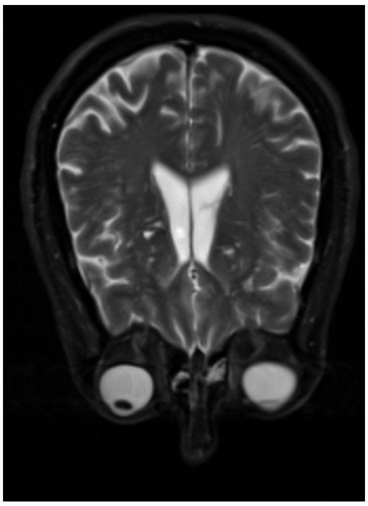

Figure 3: A MRI image from PPMI datasets

PPMI Clinical Dataset :The clinical dataset from the Parkinson's Progression Markers Initiative (PPMI) includes the following types of information:

- **Age at Visit:** Longitudinal information recorded at each visit.
- **Gait & Arm Swing (Axivity):** Motor assessment collected via wearable sensors.
- **MDS-UPDRS:** Includes four parts evaluating different aspects of Parkinson's disease:
  - Part I: Non-motor aspects of experiences of daily living
  - Part II: Movement aspects of experiences of daily living
  - Part III: Movement examination
  - Part IV: Movement complications
- **Montreal Cognitive Assessment (MoCA):** Cognitive assessment scores.
- **MRI Metadata:** Imaging-related clinical information (not raw images).
- **Biospecimen Analysis:** Biological markers and laboratory results.
- **Demographics:** Patient information including age, sex, and handedness.

5.2 Data Preprocessing

MRI Image Data: MRI images shall be normalized to range pixel values from 0 to 1. Whenever required, data augmentation techniques such as rotation, flip, and scale will can be employed to increase dataset diversity and model generalization.
Clinical/Tabular Data: Multiple tabular datasets will be merged using *Patient ID* as the key identifier. The combined dataset will be preprocessed by normalizing numerical features and encoding categorical variables. Missing values will be handled appropriately to prepare the data for subsequent modeling.

5.3 Pre-training on NTUA MRI Dataset

To leverage prior knowledge, we first employ a three-dimensional convolutional neural network (3D CNN) pretrained on the NTUA Parkinson MRI dataset, which consists of PD and non-PD cases. The use of 3D CNN allows the model to capture volumetric features across slices, including spatial and structural variations in the brain. This pretraining provides a strong initialization for subsequent fine-tuning on a larger dataset.

5.4 3D CNN

The 3D CNN consists of three major components:

- **Convolution Layer:** In 3D CNNs, the kernels slide across three dimensions (height, width, and depth) of the MRI volume to capture local voxel-level patterns. This allows the network to identify disease-related biomarkers, such as tissue density variations or abnormalities in specific brain regions. In our study, a 3D CNN will first be pre-trained on the NTUA dataset and then fine-tuned on the PPMI dataset to improve feature learning.
- **Pooling Layer:** We will use 3D pooling operations such as max pooling or average pooling to reduce the dimensionality of volumetric feature maps while retaining the most significant patterns. This helps to minimize computational cost, reduce overfitting, and improve model generalization.
- **Fully Connected Layer:** After feature extraction, the 3D feature maps are flattened and passed to fully connected layers. These layers learn high-level, non-linear relationships from the extracted features and prepare the embeddings for further fusion with clinical data. ReLU activation will be applied to introduce non-linearity and improve classification performance.

5.5 Transfer Learning on PPMI MRI Dataset

To improve model generalization and reduce training time, transfer learning will be applied by leveraging pre-trained models on external MRI datasets [10] . Specifically, a model trained on the NTUA MRI dataset will serve as the source, and its learned weights will be fine-tuned on the PPMI dataset as the target. This approach allows the model to retain previously learned spatial features while adapting to the new dataset, thereby enhancing predictive performance on Parkinson's disease classification.

5.6 Feature Extraction from MRI

From the fine-tuned 3D CNN backbone (e.g., 3D ResNet, DenseNet, or Vision Transformer), we will extract high-level volumetric embeddings that capture disease-relevant spatial and structural information. These embeddings will serve as the imaging feature representation for multimodal fusion.

5.7 Clinical Data Processing

Parallel to MRI feature extraction, the clinical data (demographics, UPDRS scores, and other biomarkers) will be processed. Clinical features will be passed through a feature engineering pipeline followed by an Excitation Network (SE-Net or MLP with squeeze-and-excitation blocks) to emphasize the most informative biomarkers. The resulting output will form the clinical embeddings.

5.8 Intermediate Fusion of Modalities

For effective multimodal integration, we will adopt **intermediate (feature-level) fusion**. Magnetic resonance imaging and clinical data will be concatenated into a unified representation. This approach will enable cross-modal interactions while preserving modality-specific characteristics. To further enhance discriminative learning, we plan to incorporate a 3D-CBAM model, which will refine the fused representation by adaptively weighting spatial and channel-wise features.

5.9 Prediction and Classification

The refined multimodal embeddings will be passed through fully connected layers, followed by a softmax activation function for stage classification based on the UPDRS. This will facilitate disease progression prediction beyond binary classification.

5.10 Explainable AI Analysis(XAI)

To ensure interpretability, (XAI) techniques will be employed. Grad-CAM will be applied to the MRI branch to highlight discriminative brain regions, while SHAP and LIME will be used for the clinical branch to rank feature importance. This interpretability will allow both clinicians and researchers to understand which features drive the model's decisions.

6 Impact Identification

6.1 Social Impact

- Early and accurate classification of PD stages can significantly improve patient quality of life by enabling timely medical interventions and slowing disease progression.
- Enhances trust in AI-assisted healthcare through explainable predictions, helping doctors, patients, and families better understand disease progression.
- Supports public health by enabling screening in at-risk populations and aiding long-term monitoring in research and clinical practice.

6.2 Economic Impact

- Reduces healthcare costs by minimizing misdiagnosis, repeated tests, and late-stage treatment expenses.

- Improves resource allocation in hospitals and research centers by prioritizing early interventions and preventive care.
- Can be scaled into cost-effective diagnostic support tools, reducing the burden on healthcare systems and making advanced PD care accessible even in resource-limited settings.

7 Application

- The proposed multimodal Parkinson's disease detection framework can assist physicians in early and accurate diagnosis by combining imaging and clinical data.
- It can be applied in hospitals and research centers to support clinical decision-making, reduce misdiagnosis, and enable timely intervention.
- The explainable AI component provides insights into key biomarkers, helping clinicians understand and trust the model's predictions.
- The framework can also be extended for screening at-risk populations and for use in longitudinal studies of disease progression.

8 Required Resources

The necessary tools to implement this project can be divided into two categories: Hardware and Software, as illustrated below:

Hardware

- Personal Computer

Software

- Jupyter Notebook with Python 3.x installed
- Google Colaboratory for GPU usage

Libraries

- Scikit-learn
- TensorFlow / Keras, PyTorch
- OpenCV
- SHAP, LIME

9 Cost Estimation

An example of a cost estimate is shown here. This can be modified according to the need.

Item	**Cost (Tk)**
Internet	3000
Paper	500
Drafting	1000
Printing	500
Binding	400
Miscellaneous	500
Total	**5900**

Table 2: Estimated project cost

10 Time Management

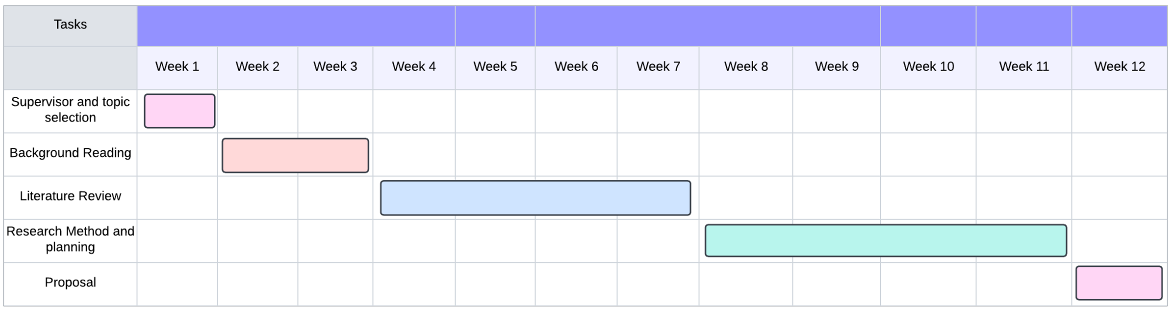

Figure 4: Time management

CSE Undergraduate Studies (CUGS) Committee

Reference :

Meeting No : Resolution No : Date :

Signature of the Student

Signature of the Supervisor

Signature of the Head of the Department

References

[1] Muhammad John Abbas, Muhammad Attique Khan, Ameer Hamza, Shrooq Alsenan, Aleesha Rehman, Jamel Baili, and Yudong Zhang. C3bam-xai: Convolutional block attention module enhanced explainable artificial intelligence-based parkinson's disease stage classification. *Cognitive Computation*, 17(3):111, 2025.

[2] Hossam Magdy Balaha, Asmaa El-Sayed Hassan, Rawan Ayman Ahmed, and Magdy Hassan Balaha. Comprehensive multimodal approach for parkinson's disease classification using artificial intelligence: insights and model explainability. *Soft Computing*, 29(3):1845–1877, 2025.

[3] Vincenzo Dentamaro, Donato Impedovo, Luca Musti, Giuseppe Pirlo, and Paolo Taurisano. Enhancing early parkinson's disease detection through multimodal deep learning and explainable ai: insights from the ppmi database. *Scientific Reports*, 14(1):20941, 2024.

[4] Adebimpe O Esan, David B Olawade, Afeez A Soladoye, Bolaji A Omodunbi, Ibrahim A Adeyanju, and Nicholas Aderinto. Explainable ai for parkinson's disease prediction: A machine learning approach with interpretable models. *Current Research in Translational Medicine*, page 103541, 2025.

[5] Parkinson's Foundation. Stages of parkinson's, 2023. Accessed: 2025-09-12.

[6] Christopher Kruse, Anna Lipinski, Malte Verheyen, Monika Balzer-Geldsetzer, Michael Wittenberg, Stefan Lorenzl, Carmen Richinger, Christian Schmotz, Lars Tönges, Dirk Woitalla, et al. Care of late-stage parkinsonism: resource utilization of the disease in five european countries. *Movement Disorders*, 39(3):571–584, 2024.

[7] Wenjia Li, Quanrui Rao, Shuying Dong, Mengyuan Zhu, Zhen Yang, Xianggeng Huang, and Guangchen Liu. Pidgn: An explainable multimodal deep learning framework for early prediction of parkinson's disease. *Journal of Neuroscience Methods*, 415:110363, 2025.

[8] Abir Noun, Racha Soubra, and Aly Chkeir. Explainable ai reveals hidden biomarkers in parkinson's brain imaging. In *2025 6th International Conference on Bioengineering for Smart Technologies (BioSMART)*, pages 1–4. IEEE, 2025.

[9] World Health Organization. Parkinson disease, August 9 2023. Accessed: 2025-09-12.

[10] Muhammad Mubinur Rahman, Hasan Murad, and Sharmistha Chanda Tista. Developing a system to detect autism spectrum disorder (asd) in children from facial

and behavioral features using multimodal approach. In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pages 986–991, 2024.